
MANIOR PORJECT

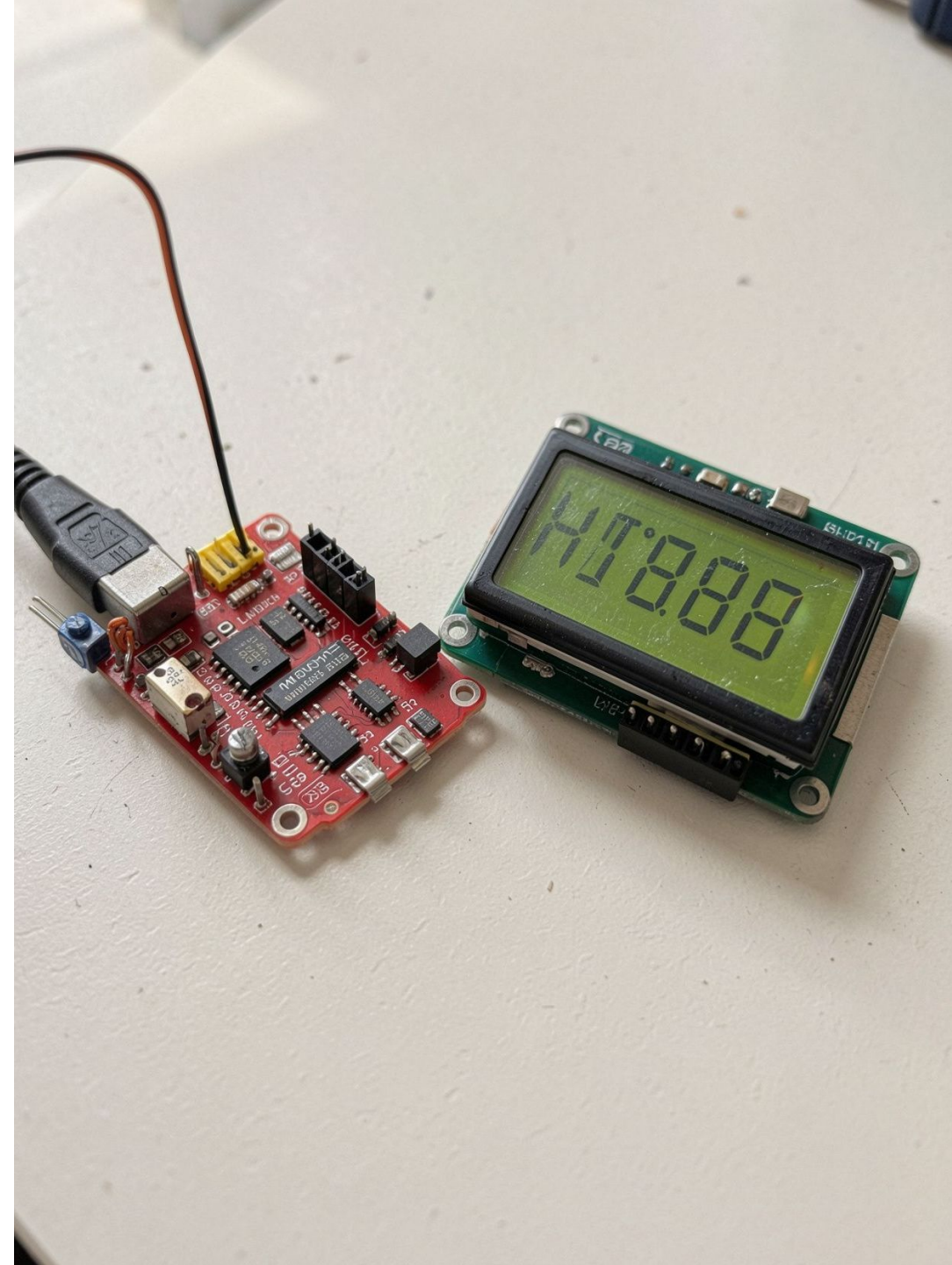
Temperature Monitoring System using Arduino.

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ELECTRONIC COMMUNICATION AND ENGINNRING

SECOND YEAR

MEENAKSHI COLLGE OF ENGINNRING



1. Project Title:

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Temperature Monitoring System using Arduino

2. Problem Statement:

Temperature monitoring is essential in industries, hospitals, and weather stations. This project focuses on designing a system that reads temperature using a sensor and displays the value in real-time. It helps in understanding sensor interfacing and data display techniques.

3. Objective:

To measure temperature using a sensor

To display temperature in °C

To understand microcontroller and sensor interfacing

To provide alert when temperature exceeds a limit

4. Tools & Technologies:

Arduino UNO / ESP32

LM35 Temperature Sensor

16x2 LCD Display

Breadboard

Jumper wires

Arduino IDE

5. Circuit Diagram:

LM35 Sensor:

VCC → 5V

GND → GND

OUT → A0

LCD Display:

VSS → GND

VDD → 5V

VO → Potentiometer

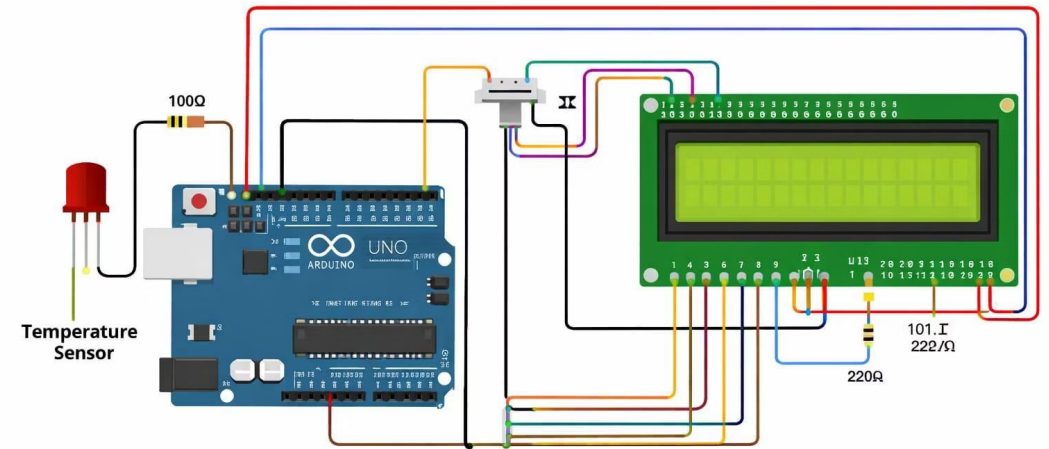
RS → Pin 7

E → Pin 6

D4-D7 → Pins 5, 4, 3, 2

6. Code:

C++



Temperature Monitoring System Circuit

#include <LiquidCrystal.h>

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
int sensorPin = A0;
float temperature;
int threshold = 40;
void setup() {
  lcd.begin(16, 2);
  lcd.print("Temp Monitor");
  delay(2000);
  lcd.clear();
}
void loop() {
  int reading = analogRead(sensorPin);
```

```
    temperature = reading * (5.0 / 1023.0) * 100;
    lcd.setCursor(0, 0);
    lcd.print("Temp: ");
    lcd.print(temperature);
    lcd.print(" C");
    if (temperature > threshold) {
      lcd.setCursor(0, 1);
      lcd.print("HIGH TEMP ALERT!");
    } else {
      lcd.setCursor(0, 1);
      lcd.print("Normal Temp");
    }
    delay(1000);
  }
```

7. Code Explanation

- 1 The sensor reads temperature as analog voltage
- 2 Arduino converts analog value into temperature
- 3 LCD displays temperature in °C
- 4 If temperature exceeds 40°C, alert message is shown

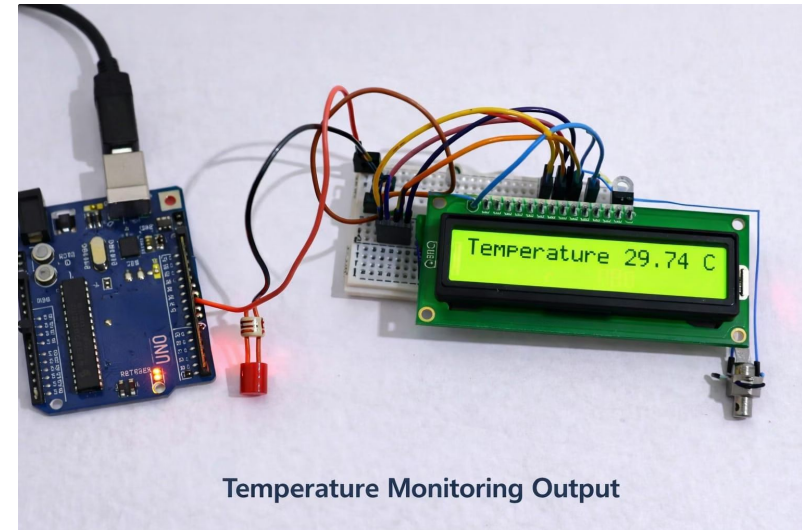
8. Output Screenshots

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Circuit setup

Temperature display on LCD

Alert message display



9. Output:

Normal Condition:

Temp: 29.74C

Normal Temp

High Temperature Alert:

Temp: 35.62

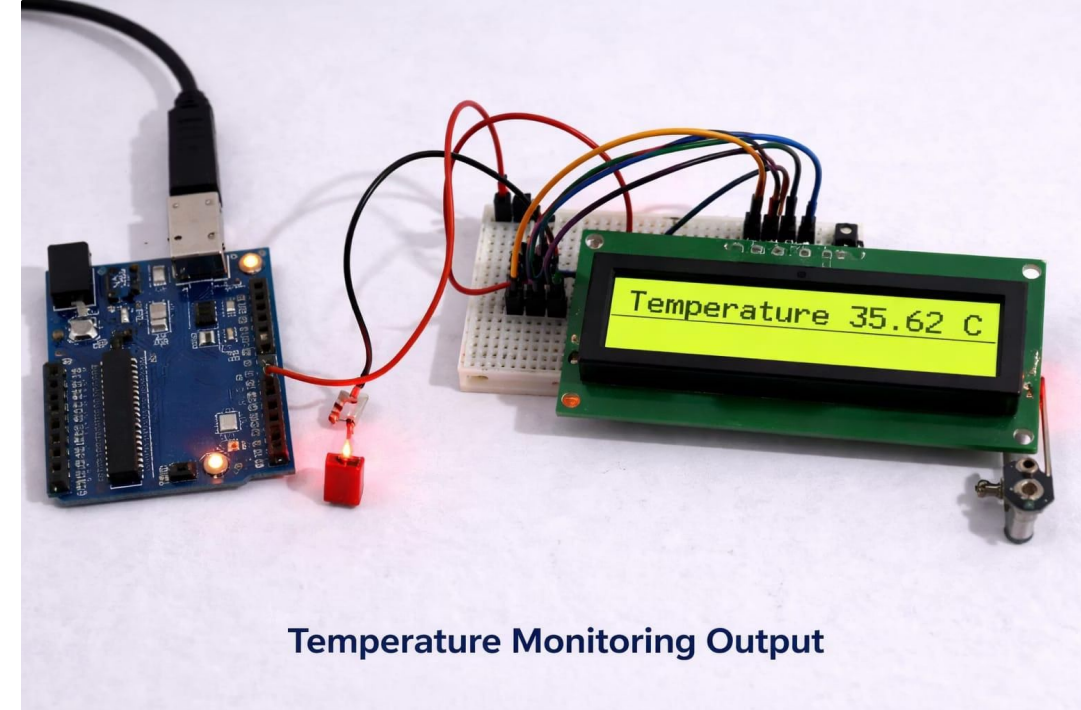
HIGH TEMP ALERT!

10. Conclusion:

The project successfully measures and displays temperature using a sensor. It also alerts the user when temperature exceeds a predefined limit. This system is useful for real-time monitoring applications.

11. Future Scope:

- Add buzzer for sound alert
- Connect to mobile using Bluetooth/WiFi
- Store temperature data
- Develop IoT-based monitoring system.



Temperature Monitoring Output